Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

Bab 1

PENDAHULUAN

The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

Bab 2

TINJAUAN PUSTAKA

Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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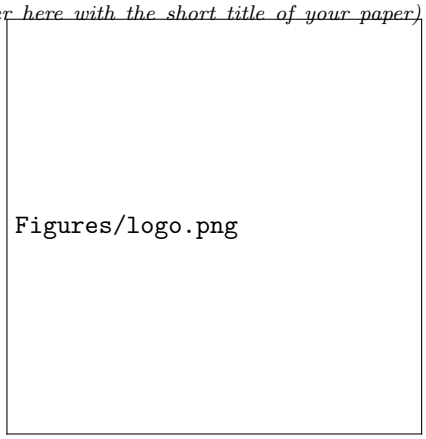
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Instructions for preparation and submission of full-papers for publication in the Turbulence Seminar

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[1]*Dept. of Something, University of Somewhere*
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Abstract. This template file provides detailed formatting instructions for preparing your full-length paper to the Proceedings of CILAMCE-2023 (XLIV Ibero-Latin American Congress on Computational Methods in Engineering). It is strongly recommended that you use the pre-defined styles of this template file, as they embed all necessary text formatting for the corresponding paragraph type. Full-length papers must be written in English.

Keywords: First keyword, Second keyword, Third keyword (up to 5 keywords)

2 Introduction

Only full-length papers that have been orally presented will be published in the congress proceedings. It is extremely important that you prepare your full-length paper in strict accordance with the text formatting of this document, which can be enforced either using the pre-defined styles of this template file or manually setting the specifications described in the next section. After the preparation of your paper, you should generate a PDF file for submission. Only PDF files will be accepted by the online submission system.

3 Format instructions

Please follow these general instructions carefully: (a) type the body of the paper in single column; (b) **use no more than 7 pages (5 pages for papers at the Research Beginners Mini-symposium)**, A4-sized, with 2.5 cm margins on all sides, and do not insert page numbers; (c) use 10pt Times New Roman throughout the body of the text, with 15pt for the paper's title 13pt for first-level headings and 10pt for second-level headings; (d) always use either exactly 13pt-spaced lines (preferred) or single-spaced lines, with justified alignment; (e) type no more than 200 words in the abstract; (f) cite references by last Name [number], and list them consecutively in the reference list by the order of citation in the text (the list should only include works that have been cited in the text); (g) provide good quality figures; (h) define all quantities, variables and symbols as soon as they first appear in the text; (i) use only SI units. We strongly encourage you to use the pre-defined styles of this template file, as they embed all text formatting described above. Full-length paper must be written in English. The general appearance of your paper should look like this document.

3.1 Writing style

Please limit your paper by writing concisely, rather than by reducing figures or tables to a size at which symbols or labels would become difficult to read.

3.2 More detailed specifications

The first page must include the paper's title (in sentence case), authors, affiliations, abstract, keywords, as well as the beginning of the text itself (i.e., the first section, which is usually the introduction). Do not insert a page break in between the keywords and the beginning of the text. Please follow strictly the line spacing defined next: either exactly 13pt-spaced lines (preferred) or single-spaced lines for the body of the text, with 20pt space before and 12pt space after first-level and second-level headings. If you use the pre-defined styles of this template, these specifications will be automatically applied.

Remark 1: Author(s) and affiliation(s). Type the authors' names in regular (plain) type, flush left, including first name, middle initial(s) and last name. Group the authors by their affiliations with a superscript number. Each name or group of names must be followed by the corresponding affiliations and emails, which should be both in italics. A 12pt space must be left between the names and affiliations. If you use the pre-defined styles of this template, these specifications will be automatically applied.

Remark 2: Abstract and keywords. Type "Abstract" in boldface, flush left, followed by a period. On the same line, type the abstract in regular (plain) format, justified alignment. The abstract should not exceed 200 words. Please pay attention to the line spacing between the authors affiliations and the abstract, as well as between the abstract and the keywords. If you use the pre-defined styles of this template, these specifications will be automatically applied. For the keywords, type "Keywords", followed by a colon, in boldface, flush left and type 3 to 5 keywords, separated by commas.

Remark 3: Headings. First-level headings must be typed in sentence letters, 13 pt boldface type,

flush left, as in Section 2 above. Use Arabic numbering (if you use the pre-defined styles of this template, automatic numbering is already built-in in the "1st Heading Cilamce-2023" style). Notice the line spacing before and after first-level headings. For second-level headings, use sentence letters, 10pt boldface type, flush left, with Arabic double numbering (if you use the pre-defined styles of this template, automatic double numbering is already defined). Notice the line spacing before and after second-level headings. Do not use third-level headings. Instead, use, at most, "Remarks" (or the like), as shown here. These shall start with the word "Remark" (or the like) in boldface italics, followed by a number and a title (optional). The text that follows must be in regular (plain) format. Notice the line spacing before and after such paragraphs.

Remark 4: Body of text. As said before, the body of the text should be typed in 10pt Times New Roman, using either exactly 13pt spaced lines (preferred) or single-spaced lines, with justified alignment. Start each paragraph with an indentation of 0.75 cm from the left margin, and allow no space between paragraphs. If you use the pre-defined styles of this template, these specifications will be automatically applied .

3.3 Equations, symbols and units

Equations must be centered, right-numbered, with numbers enclosed in parentheses and placed flush right. Allow 6 pt line spacing before and after an equation. For example:

$$q_r = -4pr^2k\frac{dT}{dr}. \tag{2.1}$$

Please pay attention to the punctuation after the equations, as equations are part of the text and must be punctuated accordingly. When referring to an equation in the text, write Eq1, except at the beginning of a sentence, wherein Equation (2.1) should be used. Please describe the notation adopted and be careful as to define all quantities, variables and symbols as soon as they first appear in the text. A nomenclature section is not necessary.

All data, including those shown in tables and figures, must be reported in SI units. Use decimal points rather than commas to indicate decimals.

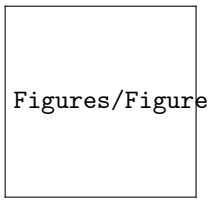
3.4 Figures and tables

Figures and tables should be inserted as close as possible to their first mentioning in the text. Embedded text and symbols must be clearly readable; avoid exceedingly small fonts. Supply good quality pictures and illustrations. Figures and tables and their captions should be centered in the text. Place figure captions below the figures and allow 12pt line spacing between both, and 20pt line spacing after the caption (i.e., before the subsequent text). Place table title above the table, leaving 20pt line spacing before the title and 6pt between the title and the table. Leave 20pt line spacing after the table (i.e., before the subsequent text). Examples are shown next:

Arabic numerals should be used in figures and tables (e.g., Figure 2.1, Figure 2, Table 2.1, Table 2, etc). Refer to them in the text as Table 2.1 and Fig1, except at the beginning of a sentence, wherein Figure 2.1 should be used.

Tabel 2.1: Coefficients in constitutive relations

Constitutive relation	Nomenclature	Value
Turbulent tensor	C_{μ}	0.09
Turbulent tensor	$C_{\mu b}$	0.69
Lateral lift	C_{L}	0.08
Virtual mass	C_{VM}	0.8



Gambar 2.1: Pressure variation along the nozzle: experimental data

When constructing graphs or plots, do not forget to label coordinates and state the corresponding units. Similarly, label all columns/rows in tables and state corresponding units whenever applicable.

3.5 Permission

You are the sole responsible for making sure that you have the right to publish everything in your paper. If you use material from a copyrighted source, or from other authors, you may need to obtain permission from the copyright holder or the respective authors. An "authorship statement" must be placed at the end of your paper, immediately before the list of references, as shown further below.

3.6 References

References should be cited in the text by Last Name [number]. For example: "In a recent work, Wriggers proposed the method of Bely". Numbers must be Arabic, enclosed in square brackets and used consecutively. References must be listed at the end of the paper, in a separate section entitled "References". They must be listed consecutively by the order of citation in the text. Type the word "References" in boldface, 13 pt Times New Roman type from the left margin, leaving 20 pt line spacing before and 12pt after. Then type the reference 9pt below it, in 9pt Times New Roman, with single-spaced lines (please do not leave blank spaces between references). See examples below. The list should only include works that are cited in the text.

4 Conclusions

Type your conclusions or closing remarks here. Please be as concise and objective as possible. Do not make a summary of the paper, but instead list the main findings and results, even if these are only partial conclusions so far.

Acknowledgements. This section should be positioned immediately after the Conclusion section. Type Acknowledgements in boldface, 10 pt Times New Roman type from left margin, leaving 20 pt line spacing before and 12pt after.

Authorship statement. This section is mandatory and should be positioned immediately before the References section. The text should be exactly as follows: The authors hereby confirm that they are the sole liable persons responsible for the authorship of this work, and that all material that has been herein included as part of the present paper is either the property (and authorship) of the authors, or has the permission of the owners to be included here.